

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	7 July 2026
Team ID	SWTID-2026-2886
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2 and The proposed technical architecture integrates MySQL, Tableau Desktop, Tableau Public, and Flask to build an interactive electricity consumption analysis system. Data is collected, processed, visualized, and published through a scalable dashboard that enables users to explore insights using dynamic filters and interactive charts.

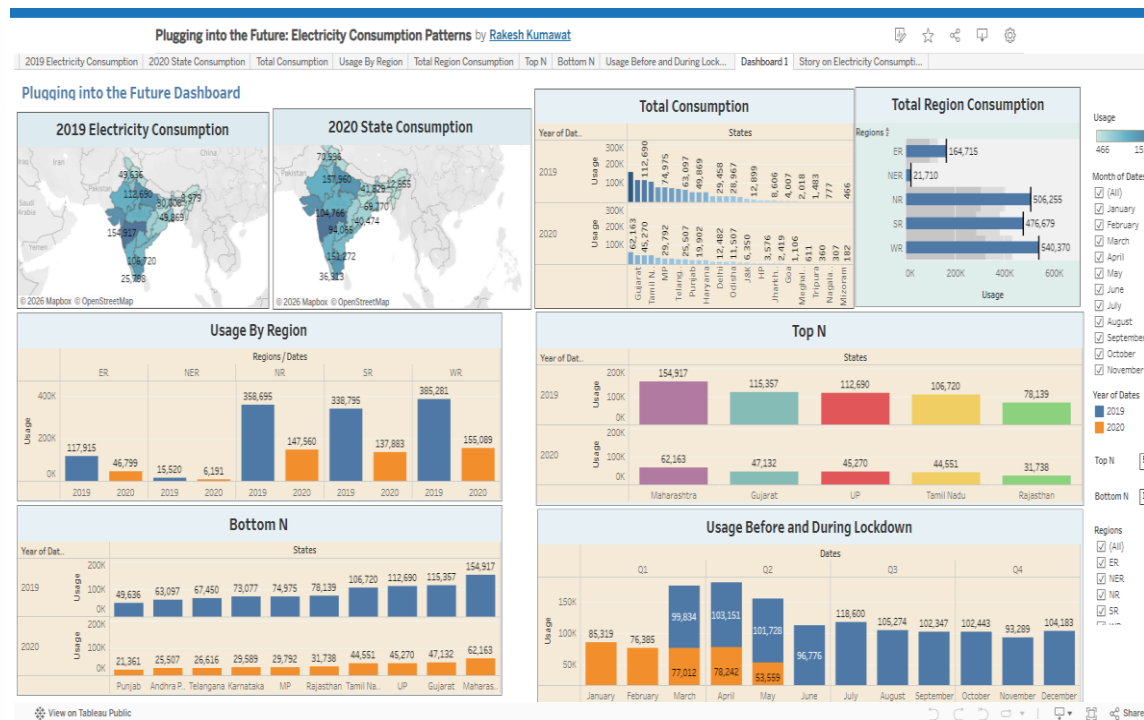
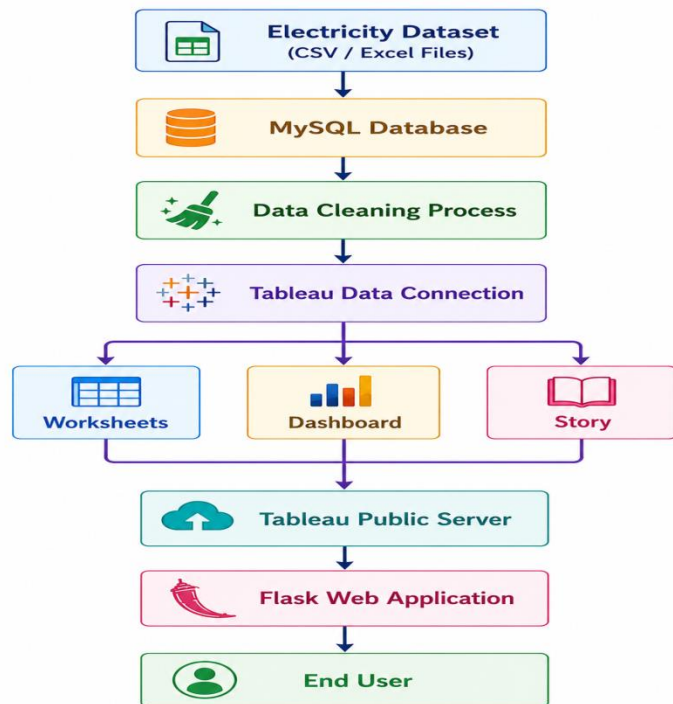


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard for users to explore electricity consumption data.	HTML, CSS

2.	Application Logic-1	Data visualization and dashboard creation	Tableau Desktop
3.	Application Logic-2	Interactive filtering and parameter-based analysis	Tableau Calculated Fields & Parameters
4.	Application Logic-3	Story creation and dashboard publishing	Tableau Story & Tableau Public
5.	Database	Stores electricity consumption records	MySQL
6.	File Storage	Stores dataset and project files	Local File System (CSV/Excel)
7.	Infrastructure (Server / Cloud)	Dashboard published online and embedded into web application	Tableau Public + Local Flask Server.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Flask framework is used for embedding the Tableau dashboard into a web interface	Flask, HTML, CSS

S.No	Characteristics	Description	Technology
2.	Security Implementations	Dashboard editing is restricted to the project owner. Public users have view-only access through Tableau Public..	Tableau Public Access Control
3.	Scalable Architecture	The solution can easily support additional years, states, and electricity datasets without major redesign.	Tableau + MySQL
4.	Availability	Dashboard is available online through Tableau Public and can also be accessed using the Flask web application.	Tableau Public
5.	Performance	Optimized dashboard using filters, parameters, and efficient visualizations for faster response.	Tableau Filters & Parameters